ALUBONETS SELF-HELP GROUP

Solution Design Document

Version 0

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Prepared for	Alubonets Self-Help Group
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1 Executive Summary

Alubonets Self-Help Group requires a secure digital platform: a public website plus a private member portal for contributions, communication, projects, meetings, and governance. This document defines what will be built, the technology it will be built on, the phases, the hosting decision the group must make, and the budget.

2 Scope — What Is Included

Everything in the client brief is in scope. Items that were vague in the brief are now stated clearly:

#	Feature	Clarified scope
1	**Public website**	Home, About, Projects, Gallery, Contact pages
2	**Member registration & login**	Registration with admin approval before access
3	**Two-Factor Authentication (2FA)**	**In scope.** OTP verification on login
4	**Member profiles**	Personal details, membership status, editable by member, managed by admin
5	**Contributions module**	Recording of member contributions, treasurer entry, member can view own history
6	**M-Pesa payments**	**In scope.** M-Pesa integration (Daraja STK Push) will be set up and ready. The group decides when to activate it for collections — the platform is built ready for it.
7	**Bulk CSV import**	**In scope.** Bulk import of existing member records and historical contributions via CSV during onboarding
8	**Welfare requests**	**In scope.** Members can submit welfare requests through the portal for review and action
9	**Announcements & events**	Admin posts announcements; events with dates visible to members
10	**Meetings & reports**	Secretary uploads meeting minutes/reports; document repository
11	**Projects**	Project listings with status, photos, and descriptions
12	**Gallery**	Photo gallery; only photos the group provides and approves will be published
13	**Financial reporting**	Treasurer dashboard, member statements, exportable reports (PDF/Word)
14	**Admin & roles**	Role-based access: Admin, Treasurer, Secretary, Member

3 Scope — What Is NOT Included (this phase)

- Loans module (deferred — can be added later as an extension)
- SMS/WhatsApp automated messaging (deferred — email notifications are included instead)
- Native mobile app (the site will be mobile-friendly/responsive; a PWA or app is a future option)
- Multi-branch support (single group only for now)

4 Proposed Technical Stack

Layer	Technology
Frontend & Application Layer	Next.js 15 (App Router)
Styling & UI	Tailwind CSS
Form Management	React Hook Form
Validation	Zod
Database ORM	Prisma
Database	Supabase PostgreSQL
Authentication	Supabase Auth
Security	Row Level Security (RLS)
Email Notifications	Resend
CSV Import Processing	PapaParse
Charts & Analytics	Chart.js
PDF Generation	pdf-lib
Word Export	docx
Payment Integration	Safaricom Daraja STK Push API

Note: this is the stack SpectreTech considers the best fit for this platform. Should any individual technology need to change during development, the system architecture below remains the same — the layered structure is independent of the specific tools, so a stack adjustment does not mean a redesign of the architecture.

5 System Architecture

Client Layer	Web browsers (mobile, tablet, desktop)
Application Layer	Next.js pages, dashboards, forms
Business Logic Layer	Server Actions, API routes, M-Pesa services, CSV processing
Data Access Layer	Prisma ORM
Data & Security Layer	Supabase PostgreSQL + Row Level Security + Auth

6 Email & Notifications

Transactional emails (registration approval, 2FA codes, announcement notifications) will be sent using **Resend**. For reliable delivery to members, the group's domain will be verified with Resend during setup — this only requires access to the domain's DNS settings, handled by SpectreTech during Phase 1. If the group does not yet own a domain, domain purchase is part of the Phase 1 setup step.

7 Hosting Decision — Option A vs Option B (Technical Comparison)

The platform is built on Next.js, Prisma, and Supabase PostgreSQL. The hosting choice affects how well this stack runs — this comparison is about technical fit.

Option A — Cloud hosting (e.g. Vercel + Supabase)

Pros

- Native fit for Next.js: server-side rendering, API routes, and server actions work out of the box.
- Supabase provides the database, authentication, and Row Level Security as a managed service with automated backups.
- Automatic scaling — performance stays stable as membership grows.
- Automatic HTTPS, CDN, and fast deployments; updates go live in minutes.
- Reliable always-on HTTPS endpoints, which M-Pesa callbacks require.

Cons

- Group depends on international providers (billing in USD).
- Deployments are managed by the developer (which SpectreTech will handle).

Option B — Shared hosting (HostAfrica, group's existing hosting)

Pros

- The group already has the account and is familiar with the provider.
- Local (Kenyan) support and billing.

Cons — important

- Typical shared hosting (cPanel/Apache/PHP) is **not designed to run a Next.js application**. Node.js support on shared plans is limited or absent; where it exists, it is restricted (memory limits, no long-running processes).

– M-Pesa callbacks require a reliable always-on HTTPS endpoint — under shared-hosting constraints these can fail or behave unreliably, as can 2FA emails and server-side rendering.
– Shared hosting databases are usually MySQL with restricted connections — the platform's Supabase PostgreSQL + RLS security model would not be available, and Prisma connection pooling can hit limits under normal member traffic.
– No automatic scaling; if the site is slow or crashes, the fix is a hosting migration anyway.

> **Decision required from the group: Option A or Option B — Option A is recommended.** The stack chosen for this platform runs natively on cloud hosting with Supabase PostgreSQL. On shared hosting the platform is at real risk of instability, especially the M-Pesa callbacks, 2FA, and the database security model.

8 Development Phases

Estimated timeline: 6–8 weeks total, commencing after deposit and scope approval.

Phase	Scope
Phase 1 — Foundation / MVP	Project setup, domain and email verification, database schema, public website, member registration with admin approval, login **with 2FA**, member profiles, admin user management, **bulk CSV import of members and historical contributions**, basic contributions recording, announcements, events, projects, gallery.
Phase 2 — Finance & Governance	Treasurer dashboard, reconciliations, financial statements, member statements, executive dashboards, secretary meeting reports, Word/PDF exports, document repository, **welfare requests**, **M-Pesa integration set up and tested (group activates for collections when ready)**. Testing, security review, deployment, training, and handover.
Phase 3 — Future Extensions (not in current budget)	Loans module, SMS/WhatsApp integration, detailed analytics, mobile app/PWA, multi-branch support.

9 Commercial Terms

Item	Detail
Project Fee	KES 20,000
Initial Deposit	50% (KES 10,000) on sign-off of this document
Balance	50% (KES 10,000) upon completion and handover
Estimated Timeline	6–8 weeks
Commencement	After deposit and scope approval

Why the revision from KES 15,000: the clarified scope now includes 2FA, M-Pesa integration setup, welfare requests, and bulk CSV data import — items that were vague or unstated in the original brief but are now confirmed in scope.

What the fee covers: the KES 20,000 covers development of the agreed Phase 1 and Phase 2 scope only.

What the fee does NOT cover: external and recurring third-party costs that may arise are billed separately and paid by the group. These include, for example: domain registration and renewal, hosting subscriptions, email service subscriptions (e.g. Resend paid tier if sending volumes require it), database/cloud service subscriptions beyond free tiers, M-Pesa Paybill/Till registration and transaction charges, and any other third-party service fees.

Scope protection: Phase 3 features (loans automation, SMS/WhatsApp notifications, advanced analytics) may be implemented as an extension depending on final requirements and third-party integration costs.

10 Risks & Mitigations

Risk	Mitigation
Scope changes during development	Formal change request process
Delayed content or approvals	Client review milestones
M-Pesa API approval delays	Integration built ready; activation independent of launch
Hosting/domain delays	Early provisioning in Phase 1
Data import quality issues	Validation using Zod and CSV checks

11 Decisions Required From the Group

1. Hosting: **Option A or Option B** (Option A recommended).
2. Official domain name: confirm existing domain or approve domain purchase.
3. M-Pesa: confirm Paybill/Till details (or confirm the group will register one) so integration can be configured.
4. Approve budget of KES 20,000 and payment terms.